Lecture 7. Estimation of Basic Quantities for Other Sampling Schemes

Mean and Quantiles; Left Truncation; Left Censoring; Right Truncation; Turnbull

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1 Estimating the Mean Survival Time

1.1 Mean as the area under the curve

Since $\mu = \mathrm{E}(X) = \int_0^\infty S(t)\,\mathrm{d}t$, the natural estimator is $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$. Problem: if the last observation is censored, $\hat{S}(t)$ stays flat (never reaches 0) and the integral diverges. The right tail can be completed, but the estimate of the mean then depends on which completion is used.

1.2 First solution: Efron's completion

Setting $\hat{S}(t) = 0$ for $t > t_{\max}$ (Efron) gives $\hat{\mu} = \int_0^{t_{\max}} \hat{S}(t)\,\mathrm{d}t$. Downsides: it assumes all survivors at $t_{\max}$ die immediately afterwards (biased downward), and $t_{\max}$ varies from sample to sample.

1.3 Second solution: restricted mean

Theorem 7.1: Restricted mean

For a pre-specified τ, estimate $\mu_\tau = \mathrm{E}[\min(X,\tau)] = \int_0^\tau S(t)\,\mathrm{d}t$ by

$$\hat{\mu}_\tau = \int_0^\tau \hat{S}(t)\,\mathrm{d}t, \qquad \hat{V}(\hat{\mu}_\tau) = \sum_{j=1}^{D} \Big[\int_{t_j}^\tau \hat{S}(t)\,\mathrm{d}t\Big]^2 \frac{d_j}{Y_j(Y_j-d_j)},$$

with $100(1-\alpha)\%$ CI $\hat{\mu}_\tau \pm Z_{1-\alpha/2}\sqrt{\hat{V}(\hat{\mu}_\tau)}$. (The sum runs over death times $t_j \le \tau$.)

Remark 7.2: Why the weights $\int_{t_j}^\tau \hat{S}$

A perturbation of the hazard estimate at t_j rescales $\hat{S}(t)$ for every $t \ge t_j$, hence changes the area to the right of t_j only. By the delta method, the contribution of $\hat{h}_j$ to $\mathrm{Var}(\hat{\mu}_\tau)$ is (area right of $t_j)^2\times$ Greenwood's term $d_j/[Y_j(Y_j-d_j)]$.

Since $\hat{S}$ is a step function, $\hat{\mu}_\tau$ is a sum of rectangles: $\hat{\mu}_\tau = \sum_k \hat{S}(s_k)(s_{k+1}-s_k)$ over consecutive jump points s_k in $[0,\tau]$.

Example 7.3: 6-MP group (data of Lecture 5) [Supplement]

KM: $\hat{S} = 0.857, 0.807, 0.753, 0.690, 0.627, 0.538, 0.448$ at $t_j = 6,7,10,13,16,22,23$ ($Y_j = 21,17,15,12,11,7,6$). With $\tau = 35$ (the largest, censored, time):

$$\begin{aligned}\hat{\mu}_{35} &= 6(1) + 1(0.857) + 3(0.807) + 3(0.753) + 3(0.690)\\ &\quad + 6(0.627) + 1(0.538) + 12(0.448) \approx 23.29 \text{ weeks},\\ \hat{V}(\hat{\mu}_{35}) &\approx 7.99, \quad \mathrm{SE} \approx 2.83, \quad 95\%\ \mathrm{CI} \approx (17.75,\ 28.83).\end{aligned}$$

2 Quantiles and Their Confidence Intervals

Recall $x_p = \inf\{t : S(t) \le 1-p\}$ ($p = 0.5$: median). The estimator is $\hat{x}_p = \inf\{t : \hat{S}(t) \le 1-p\}$; its variance depends on the density $f(x_p)$, which is hard to estimate. Instead, CIs are obtained by **inverting** the pointwise CIs for $S(t)$.

Theorem 7.4: CIs for x_p

A $100(1-\alpha)\%$ CI for x_p is the set of all t satisfying

$$\text{linear:} \quad -Z_{1-\alpha/2} \le \frac{\hat{S}(t)-(1-p)}{\hat{V}^{1/2}[\hat{S}(t)]} \le Z_{1-\alpha/2},$$

$$\text{log-log:} \quad \left|\frac{[\log\{-\log\hat{S}(t)\} - \log\{-\log(1-p)\}]\hat{S}(t)\log\hat{S}(t)}{\hat{V}^{1/2}[\hat{S}(t)]}\right| \le Z_{1-\alpha/2},$$

$$\text{arcsine:} \quad \left|\frac{2\{\arcsin\sqrt{\hat{S}(t)} - \arcsin\sqrt{1-p}\}\{\hat{S}(t)(1-\hat{S}(t))\}^{1/2}}{\hat{V}^{1/2}[\hat{S}(t)]}\right| \le Z_{1-\alpha/2}.$$

Derivation: where these come from

Each is a Wald test of $H_0 : S(t) = 1 - p$ on the transformed scale: $|g(\hat{S}) - g(1-p)|/(|g'(\hat{S})|\,\mathrm{SE}(\hat{S})) \le Z$. For log-log, $1/|g'(\hat{S})| = |\hat{S}\log\hat{S}|$; for arcsine-square-root, $1/|g'(\hat{S})| = 2\sqrt{\hat{S}(1-\hat{S})}$. The CI collects all t at which H_0 is not rejected.

Example 7.5: Median for the 6-MP group

$\hat{S}(22) = 0.538 > 0.5 \ge \hat{S}(23) = 0.448$, so $\hat{x}_{0.5} = 23$. Checking the criteria at each death time, all three methods are satisfied from $t = 13$ onwards ($\hat{S}(13) = 0.690$, SE 0.107) but not at $t \le 10$. Since $\hat{S}$ never drops far enough below 0.5 to violate the criterion again, the 95% CI for the median is $[13, \infty)$: the upper limit cannot be determined.

3 Right-Censored and Left-Truncated Data

Definition 7.6: Data and modified risk set

Data $\{(L_i, T_i, \delta_i)\}$: L_i = age at entering the study, T_i = death or censoring time. At death times $t_1 < \cdots < t_D$ let d_j = number of deaths and

$$Y_j = \sum_{i=1}^{n} I(L_i \le t_j \le T_i),$$

the number who have *entered* and are still under observation at t_j.

With this modified Y_j, all estimators of K&M Sections 4.2–4.4 (KM, Nelson–Aalen, Greenwood, CIs, bands) apply unchanged.

Remark 7.7: What is being estimated

Subjects are only seen after their entry time, so the estimators target *conditional* quantities:

$$S_L(t) = P(X > t \mid X > L), \qquad H_L(t) = -\ln P(X > t) + \ln P(X > L), \qquad H_L'(t) = h(t).$$

The hazard itself is unaffected by conditioning, which is why d_j/Y_j still estimates it.

Example 7.8: Channing House

Residents of the Channing House retirement center (California): 97 males and 365 females; truncation times are the ages (months) at entry. The number at risk is small at young ages, rises to a peak around 950–1000 months, then falls.

- Males: the first subject entered at 751 and died at 777; the second entered at 759 and died at 781; the third appeared only at 782.
- Plain KM conditional on the youngest entry: at 777, $Y = 2$, $d = 1 \Rightarrow$ factor $1/2$; at 781, $Y = 1$, $d = 1 \Rightarrow$ factor 0. So $\hat{S}_L(t) = 0$ for all $t \ge 781$ — the estimate is destroyed by the tiny early risk set.
- Remedy: estimate $S_a(t) = P(X > t \mid X \ge a)$ for a chosen age a where the risk set is

reasonably large:

$$\hat{S}_a(t) = \prod_{a \le t_j \le t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}_a(t)] = \hat{S}_a(t)^2 \sum_{a \le t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)},$$

$$\hat{H}_a(t) = \sum_{a \le t_j \le t} \frac{d_j}{Y_j}, \qquad t \ge a.$$

4 Left-Censored Data: Reversing the Time Scale

X = failure time, C = left censoring time; observe $T = \max(X, C)$, $\delta = I(T = X) = I(X \ge C)$. How to estimate $S(u) = P(X > u)$?

Theorem 7.9: Reversed time-scale trick

Pick $\tau > \max_i T_i$. Then $\tau - X_i$ is *right* censored by $\tau - C_i$:

$$\tau - T_i = \min(\tau - X_i,\ \tau - C_i), \qquad \delta_i = I(\tau - X_i \le \tau - C_i).$$

Let $G(u) = P(\tau - X > u)$. Apply the usual KM to $\{(\tau - T_i, \delta_i)\}$ to get $\hat{G}$, and set

$$\hat{S}(u) = 1 - \hat{G}(\tau - u), \qquad \mathrm{Var}[\hat{S}(u)] = \mathrm{Var}[\hat{G}(\tau - u)].$$

CIs and bands are built the same way.

Why [Supplement]. $\max(X, C) = \tau - \min(\tau - X, \tau - C)$, and $X \ge C \iff \tau - X \le \tau - C$. Also $G(\tau - u) = P(\tau - X > \tau - u) = P(X < u)$, so $S(u) = 1 - G(\tau - u)$ (for continuous X). ■

5 Right-Truncated Data

We observe $\{(R_i, X_i)\}$ sampled conditionally on $X_i < R_i$. Reverse the time scale with $\tau > \max_i R_i$: then $\tau - X_i$ is *left truncated* by $\tau - R_i$ (since $X_i < R_i \iff \tau - X_i > \tau - R_i$). Use the standard left-truncation method to estimate the conditional distribution of $\tau - X_i$ given $\tau - X_i > a$.

Definition 7.10: Choice $a = \tau - \tau^*$, $\tau^* = \max_i R_i$

$$G^*(t) = P(\tau - X_i \ge t \mid \tau - X_i \ge \tau - \tau^*) = P(\tau - X_i \ge t \mid X_i \le \tau^*),$$
$$S^*(t) = P(X_i > t \mid X_i \le \tau^*) = 1 - G^*(\tau - t).$$

Theorem 7.11: KM on the reversed scale

With $\tau - t_j$ the D distinct failure times on the reversed scale,

$$\hat{G}^*(t) = \prod_{\tau - \tau^* \le \tau - t_j \le t} \Big(1 - \frac{d_j}{Y_j}\Big),$$

$$d_j = \#\{i : \tau - X_i = \tau - t_j\}, \quad Y_j = \#\{i : \tau - X_i \ge \tau - t_j \ge \tau - R_i\},$$

and $\hat{S}^*(t) = 1 - \hat{G}^*(\tau - t)$. Inference for S^* uses the properties of $\hat{G}^*$ for LT data.

Only the conditional survival given $X \le \tau^*$ is estimable: nothing in the data speaks about $X > \tau^*$.

Remark 7.12: Quick review

KM and NA were developed for a single, possibly right-censored failure time with independent censoring. They extend directly to LT + RC data (modified risk set), and to left-censored or right-truncated data by reversing the time scale.

6 Other Sampling Schemes

Interval censoring, double censoring, and their combinations with left truncation have no simple closed-form estimator; Turnbull's iterative methods for these cases are covered in Lecture 8.

7 Selected exercises

Motivation. The restricted mean and the left-truncation risk set were used above without full justification; these exercises supply the identity behind μ_τ, the delta-method derivation of its variance, and the reason why d_j/Y_j with the modified risk set still estimates the hazard.

Exercise 7.1 (The restricted mean)

Let $X \ge 0$ have survival function S and let $\tau > 0$.
(a) Show that $\mathrm{E}[\min(X,\tau)] = \int_0^\tau S(t)\,\mathrm{d}t$.
(b) Without censoring and with $\tau \ge X_{(n)}$, show that $\hat{\mu}_\tau = \bar{X}$.

Solution [Supplement].

(a) Write $\min(x,\tau) = \int_0^\tau I(t < x)\,\mathrm{d}t$ for $x \ge 0$. By Tonelli (the integrand is nonnegative),

$$\mathrm{E}[\min(X,\tau)] = \int_0^\tau P(X > t)\,\mathrm{d}t = \int_0^\tau S(t)\,\mathrm{d}t.$$

(b) By Exercise 5.1(a), $\hat{S}(t) = n^{-1}\sum_i I(X_i > t)$, which vanishes for $t \ge X_{(n)}$. Hence

$$\hat{\mu}_\tau = \frac{1}{n}\sum_{i=1}^n \int_0^\tau I(t < X_i)\,\mathrm{d}t = \frac{1}{n}\sum_{i=1}^n \min(X_i,\tau) = \bar{X}.$$

So the restricted mean is the natural censored-data version of the sample mean. ■

Exercise 7.2 (Variance of the restricted mean)

Derive the variance estimator $\hat{V}(\hat{\mu}_\tau)$ of this lecture by the delta method, under the same approximations as Greenwood's formula (Lecture 5): the $\hat{h}_j = d_j/Y_j$ are approximately uncorrelated with $\mathrm{Var}(\hat{h}_j) \approx \hat{h}_j(1-\hat{h}_j)/Y_j$; assume $Y_j > d_j$ for $t_j \le \tau$.

Solution [Supplement].

Step 1 (dependence on $\hat{h}_j$). $\hat{S}(t) = \prod_{t_j \le t}(1-\hat{h}_j)$, so for $t_j \le \tau$,

$$\frac{\partial \hat{S}(t)}{\partial \hat{h}_j} = -\frac{\hat{S}(t)}{1-\hat{h}_j}\,I(t \ge t_j).$$

Step 2 (gradient of $\hat{\mu}_\tau$). Differentiating under the (finite-sum) integral,

$$\frac{\partial \hat{\mu}_\tau}{\partial \hat{h}_j} = -\frac{1}{1-\hat{h}_j}\int_{t_j}^{\tau} \hat{S}(t)\,\mathrm{d}t =: -\frac{A_j}{1-\hat{h}_j}.$$

Step 3 (delta method). With uncorrelated $\hat{h}_j$,

$$\mathrm{Var}(\hat{\mu}_\tau) \approx \sum_{t_j \le \tau} \frac{A_j^2}{(1-\hat{h}_j)^2} \cdot \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j} = \sum_{t_j \le \tau} A_j^2 \frac{\hat{h}_j}{Y_j(1-\hat{h}_j)} = \sum_{t_j \le \tau} A_j^2 \frac{d_j}{Y_j(Y_j-d_j)},$$

which is $\hat{V}(\hat{\mu}_\tau)$. The same computation with the functional $\hat{S}(t)$ in place of $\hat{\mu}_\tau$ (gradient $-\hat{S}(t)/(1-\hat{h}_j)$) gives Greenwood's formula; for the area, $\hat{S}(t)$ is replaced by the area A_j to the right of t_j, as described in the Remark above. ■

Exercise 7.3 (Why the modified risk set works under left truncation)

Let X be discrete with support $\{x_j\}$ and hazard $h(x_j) = P(X = x_j)/P(X \ge x_j)$. Let the entry time L and censoring time C be such that X, L, C are mutually independent; a subject is sampled only if $L \le X$ and then yields (L, T, δ) with $T = \min(X, C)$, $\delta = I(X \le C)$. Show that, for every x_j with $P(L \le x_j \le T) > 0$,

$$P(T = x_j,\ \delta = 1 \mid L \le x_j \le T,\ \text{sampled}) = h(x_j).$$

Solution [Supplement].

Step 1 (sampling drops out). $\{L \le x_j \le T\}$ implies $L \le x_j \le X$, so it is contained in the sampling event $\{L \le X\}$. Hence conditioning on "sampled" in addition changes nothing:

$$P(\cdot \mid L \le x_j \le T,\ \text{sampled}) = P(\cdot \mid L \le x_j \le T).$$

Step 2 (independence).

$$\frac{P(X = x_j,\ C \ge x_j,\ L \le x_j)}{P(X \ge x_j,\ C \ge x_j,\ L \le x_j)} = \frac{P(X = x_j)P(C \ge x_j)P(L \le x_j)}{P(X \ge x_j)P(C \ge x_j)P(L \le x_j)} = h(x_j).$$

The numerator is $P(T = x_j, \delta = 1,\ L \le x_j \le T)$, and the denominator is $P(L \le x_j \le T)$.

So among subjects with $L_i \le t_j \le T_i$ (exactly the modified risk set Y_j) the chance of an observed death at t_j is $h(t_j)$, and d_j/Y_j estimates it. The product of $1-h$ over $x_j \ge a$ gives $P(X > t \mid X \ge a)$, which is why only conditional survival is estimable. ■